

Introduction to Bioinformatics

RNA Structure and Transcription

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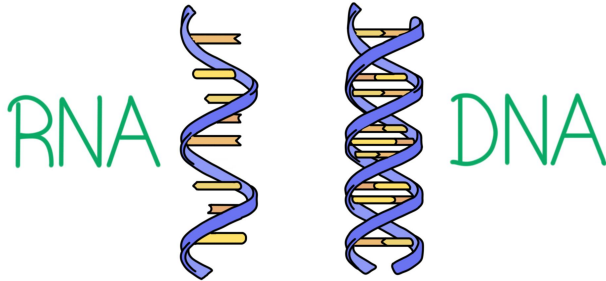
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Presentation Overview

- ① RNA Structure and Chemistry
- ② Transcription: From DNA to RNA
- ③ Types of RNA and Gene Regulation

The RNA World Hypothesis — Origin of Modern Biology



Concept:

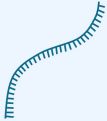
- Early life may have relied on RNA both for **information storage** and **catalysis**
- RNA molecules capable of self-replication could have preceded DNA and proteins
- Modern examples: ribozymes, ribosomal RNA catalysis

Significance

RNA's dual ability to store genetic information and catalyze reactions supports its central role in the origin of life.

Structured vs. Unstructured RNA Molecules

mRNA



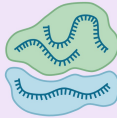
Encodes proteins

tRNA



Carries amino acids

rRNA



Forms the ribosome

Some RNA regions remain **unstructured**, like parts of **mRNA**.

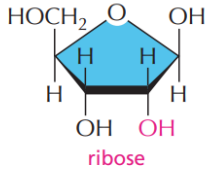
Others fold into **complex 3D structures**, such as:

- **tRNA** (adaptor in translation)
- **Ribozymes** (catalytic RNAs)
- **rRNA** (structural core of ribosome)

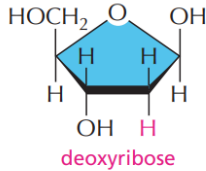
Structure reflects RNA's diverse cellular roles.

RNA Differs from DNA in Two Key Ways

(A) SUGAR DIFFERENCES

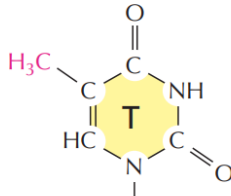
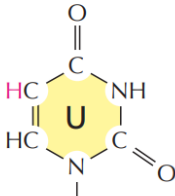


used in RNA



used in DNA

(B) BASE DIFFERENCES

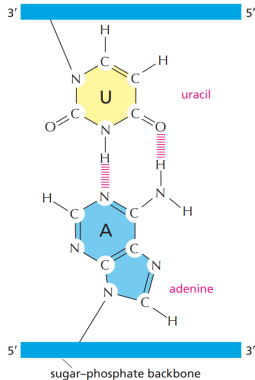


1. Sugar: RNA contains **ribose** (extra -OH group)

2. Base: RNA uses **uracil (U)** instead of thymine (T)

Both linked by phosphodiester bonds

Uracil Pairs with Adenine Like Thymine



U forms hydrogen bonds with A, just like T does in DNA

U lacks the methyl group ($-\text{CH}_3$) present in T

Same base-pairing rules apply to RNA

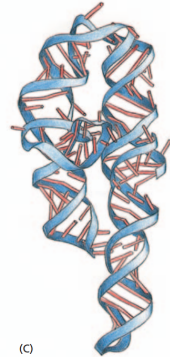
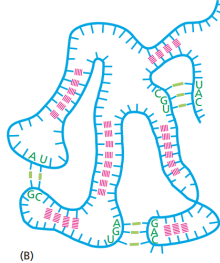
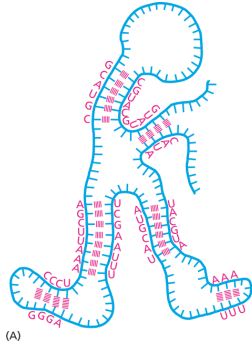
RNA is Single-Stranded — DNA is Double-Stranded

Key Structural Difference

- DNA: always double helix
- RNA: single-stranded → can fold into complex 3D shapes
- This allows RNA to have structural, catalytic, and regulatory roles

RNA's ability to fold enables diverse cellular functions beyond just carrying information

RNA Folds Into Complex Structures via Base-Pairing



RNA Secondary Structure — Key Features

How RNA Folds

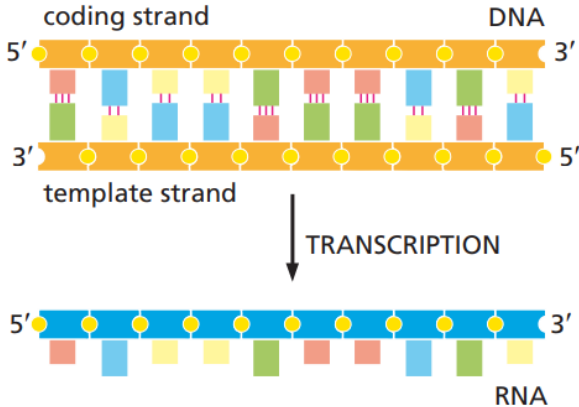
- RNA can base-pair with itself at complementary regions
- Forms hairpins, loops, and double-helical stems
- Uses conventional base pairs (G-C, A-U) and non-conventional pairs (e.g., G-U)
- 3D structure is determined by nucleotide sequence

Functional Importance

This folding capability allows RNA to function as:

- Structural scaffold (e.g., ribosomal RNA)
- Catalytic molecule (ribozymes)
- Regulatory element (e.g., microRNA)

Transcription: RNA Complementary to DNA



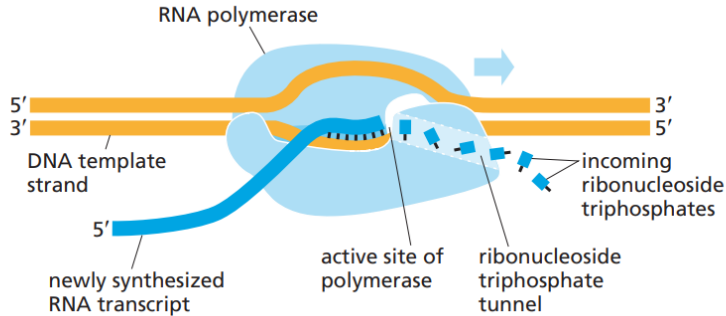
One DNA strand serves as **template**

RNA polymerase adds ribonucleotides 5'→3'

RNA sequence is complementary to template strand

RNA is single-stranded and released

RNA Polymerase Transcribes DNA into RNA



RNA Polymerase Mechanism

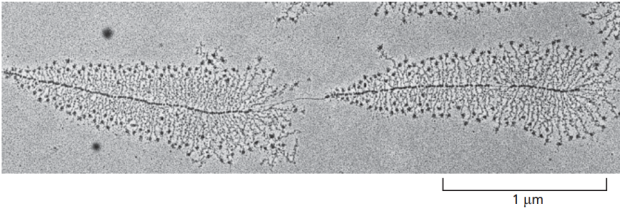
Step-by-Step Process

- 1 **DNA Unwinding:** RNA polymerase unwinds the DNA double helix
- 2 **Template Recognition:** Uses one strand as template (read 3'→5')
- 3 **RNA Synthesis:** Adds ribonucleotides 5'→3' direction
- 4 **DNA Re-formation:** DNA double helix re-forms behind the polymerase

Key Differences from DNA Replication

- RNA polymerase does NOT require a primer
- Only one DNA strand is used as template
- RNA remains single-stranded
- Lower fidelity: ~ 1 error per 10^4 nucleotides

Multiple Polymerases Transcribe Simultaneously



Many polymerases on same gene

Growing RNA transcripts visible as threads

Enables rapid mRNA production

Ribosomal proteins assemble on rRNA as it's synthesized

Cells Produce Various Types of RNA

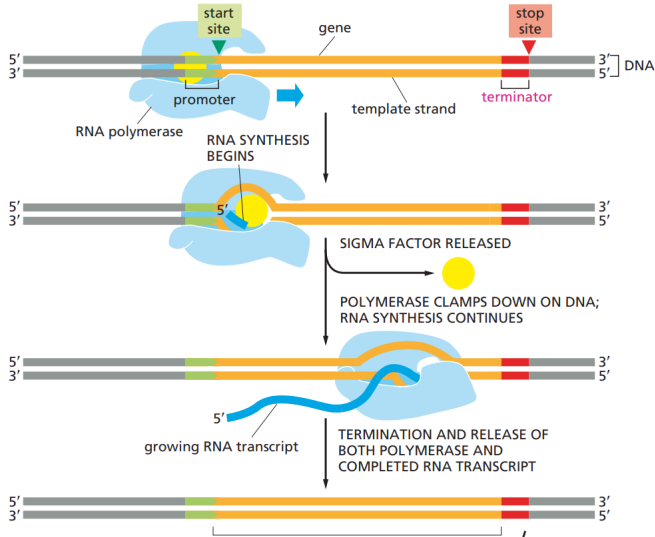
Major RNA Types

- **mRNA** (messenger RNA): encodes proteins
- **rRNA** (ribosomal RNA): structural/catalytic core of ribosomes
- **tRNA** (transfer RNA): amino acid adaptors during translation
- **miRNA** (microRNA): gene regulation
- Other noncoding RNAs: splicing, telomere maintenance, etc.

Note

Not all genes encode proteins — many produce functional RNA molecules!

Promoters and Terminators Signal Start and Stop



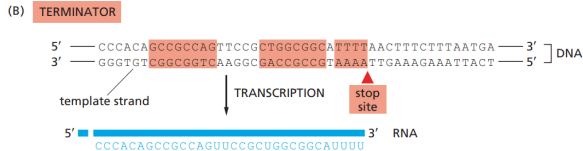
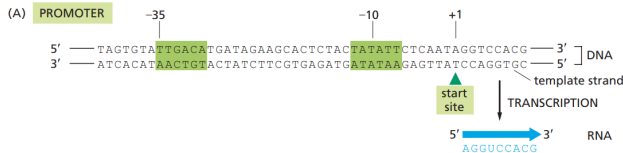
Promoter: RNA polymerase binds here

Sigma factor recognizes promoter

Terminator: signals transcription end

Polymerase releases RNA and DNA

Bacterial Promoters Have Conserved Sequences

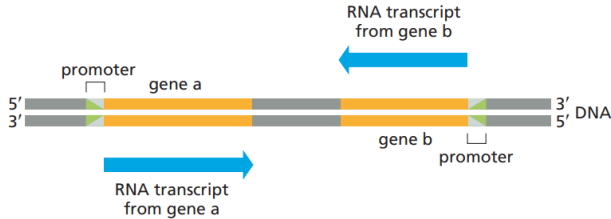


Sequences at -10 and -35 positions

Promoter polarity determines which strand is template

Only one reading frame is correct

Promoter Orientation Determines Template Strand

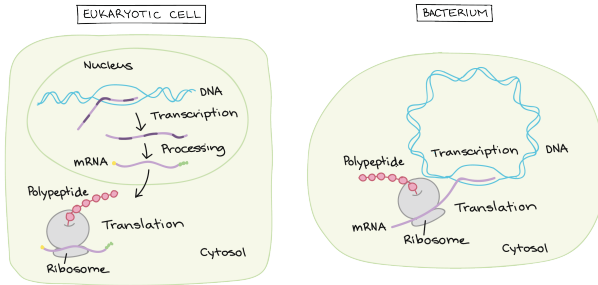


Different genes use different strands

Promoter orientation controls direction

RNA polymerase always moves 3'→5' on template

RNA Processing in Eukaryotes: From pre-mRNA to mRNA



Eukaryotic RNAs undergo several modifications before leaving the nucleus:

- **5' Capping:** addition of a modified guanine nucleotide
- **Splicing:** removal of noncoding *introns* and joining of *exons*
- **3' Polyadenylation:** addition of poly(A) tail

These steps stabilize RNA and prepare it for translation in the cytoplasm.

Summary — RNA Structure and Transcription

Key Concepts

- RNA differs from DNA by sugar (ribose) and base (uracil)
- RNA is single-stranded and folds into functional 3D shapes
- Transcription uses RNA polymerase to synthesize RNA from DNA template
- Promoters and terminators define transcription boundaries
- In eukaryotes, pre-mRNA undergoes capping, splicing, and polyadenylation

Takeaway

RNA is more than a messenger — it plays structural, catalytic, and regulatory roles essential to gene expression and cellular control.

References



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